

Assignment - 04

Life Skills - II (Aptitude)

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Q1. What is the number of permutations of the word "BOOK"?

- (a) 12 (b) 24 (c) 120 (d) 720

Sol.

BOOK $\rightarrow$ Total 4 letters but letter 'O' appears twice.

So, Permutation of 4 objects

$$P_4 = 4! = 4 \times 3 \times 2 \times 1 = 24$$

and permutation of twice letter 'O' $P_2 = 2! = 2 \times 1 = 2$

So, $P = \frac{24}{2} = 12$

— (a)

Q2. How many ways can you arrange the letters in the word "MATH"?

- (a) 12 (b) 24 (c) 72

Sol.

"MATH"

Total letter (n) = 4

So, we use this formula to calculate permutations:

$$\frac{n!}{(n-k)!}$$

Here n = number of items available

k = no. of items you want to select

So

$$\frac{n!}{(n-k)!} = \frac{4!}{(4-4)!} = \frac{4 \times 3 \times 2 \times 1}{0!} = \frac{24}{1}$$

$$= 24$$

— (b)